

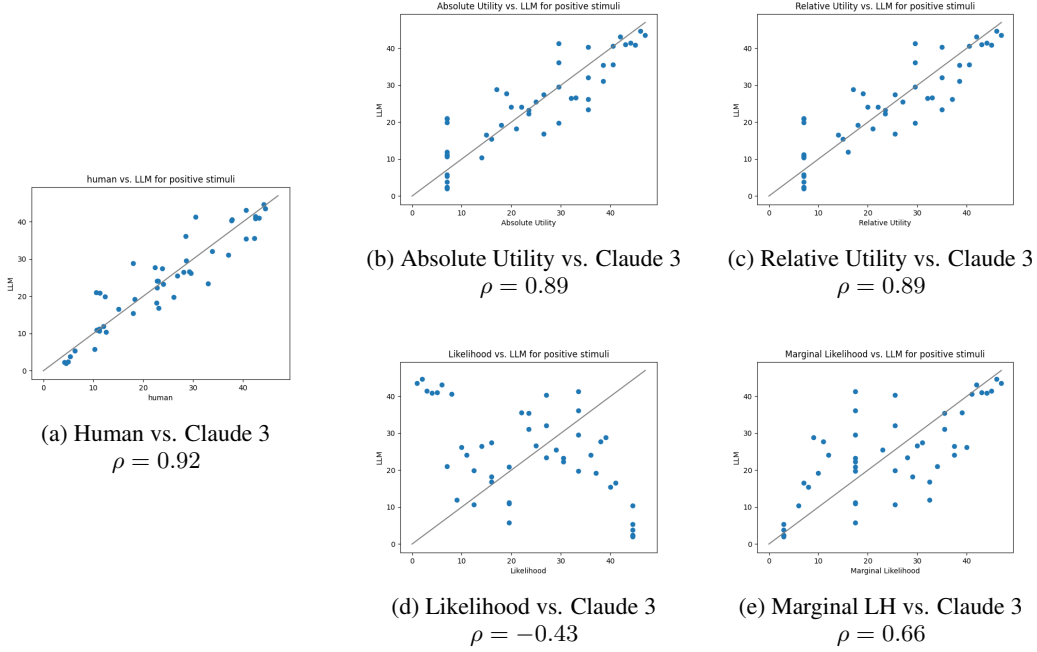


Figure 5: Comparing Claude 3 Opus CoT rankings (y-coordinates) to humans and four theoretical decision-making models (x-coordinates) in positive setting.

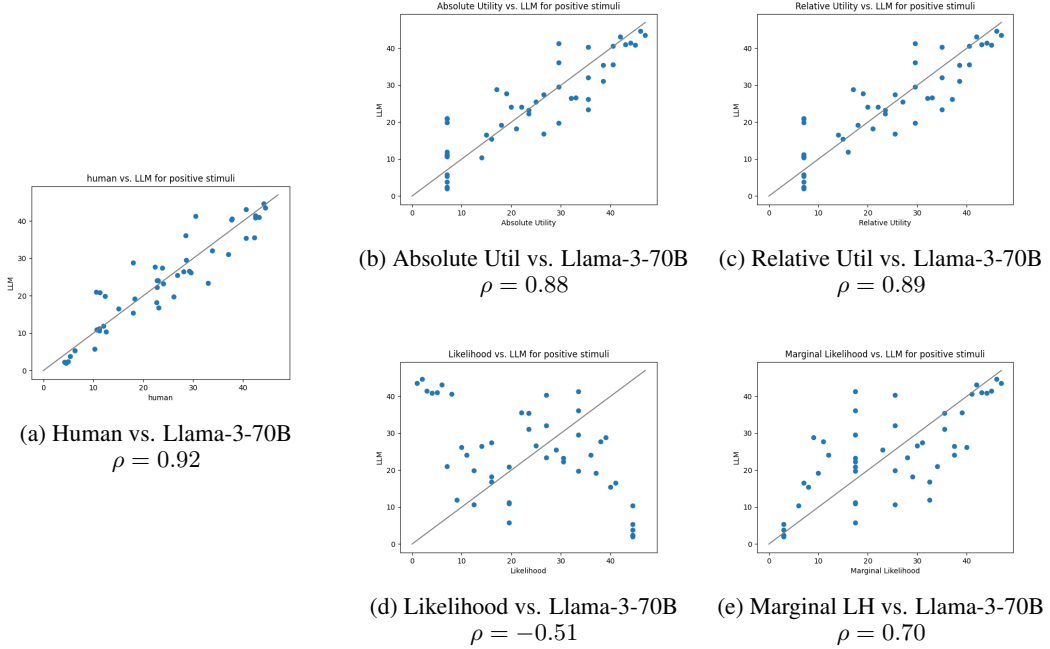


Figure 6: Comparing Llama-3-70B CoT rankings (y-coordinates) to humans and four theoretical decision-making models (x-coordinates) in negative setting.

Table 13: Comparison of Zero-shot vs. Human and CoT vs. Human across temperature settings for GPT-4o.

Zero-shot vs. Human		CoT vs. Human	
Temp	Spearman	Temp	Spearman
0.0	0.533	0.0	0.540
0.5	0.381	0.5	0.515
1.0	0.341	1.0	0.518
1.5	0.489	1.5	0.528
2.0	<i>model unable</i>	2.0	<i>model unable</i>

Table 14: Comparison of Zero-shot vs. Max Expected Value and CoT vs. Max Expected Value.

Zero-shot vs. Max Expected Value			CoT vs. Max Expected Value		
Demographic	Spearman	p-value	Demographic	Spearman	p-value
Woman 18-35	0.289	9.44E-21	Woman 18-35	0.970	<1E-150
Woman 45-60	0.313	4.23E-24	Woman 45-60	0.972	<1E-150
Woman 65-85	0.285	9.91E-21	Woman 65-85	0.969	<1E-150
Man 18-35	0.335	2.50E-27	Man 18-35	0.971	<1E-150
Man 45-60	0.308	3.37E-23	Man 45-60	0.965	<1E-150
Man 65-85	0.348	3.98E-29	Man 65-85	0.964	<1E-150
Monkey	0.208	3.49E-13	Monkey	0.867	7.37E-123

G INVERSE MODELING PROMPTS

Example inverse modeling prompts for zero-shot and CoT are shown in Tables 16 and 17. First, the context of the experiment is introduced, then the choices are listed, and lastly the LLM is asked to reply with which comparison more strongly suggests that the decision-maker prefers a certain target item.

H 47 DECISIONS USED IN INVERSE DECISION-MAKING EXPERIMENT

We provide a list of the 47 decisions used in the inverse decision-making experiment of Jern et al. (2017) in Table 18. Columns represent options, and letters represent items with the options. Based on the context, letters were replaced with colored candies or numbered electric shocks. Participants ranked these decisions by their strength in suggesting that the decision-maker preferred item x over the other items.

Table 15: Comparison of Zero-shot vs. Human and CoT vs. Human.

Zero-shot vs. Human			CoT vs. Human		
Demographic	Spearman	p-value	Demographic	Spearman	p-value
Woman 18-35	0.529	2.75E-73	Woman 18-35	0.549	9.23E-80
Woman 45-60	0.519	6.07E-70	Woman 45-60	0.546	3.54E-79
Woman 65-85	0.520	2.75E-73	Woman 65-85	0.550	3.70E-80
Man 18-35	0.499	4.18E-64	Man 18-35	0.528	2.10E-78
Man 45-60	0.530	1.43E-73	Man 45-60	0.539	1.34E-76
Man 65-85	0.508	8.54E-67	Man 65-85	0.531	3.57E-74
Monkey	0.482	2.18E-59	Monkey	0.630	8.47E-112

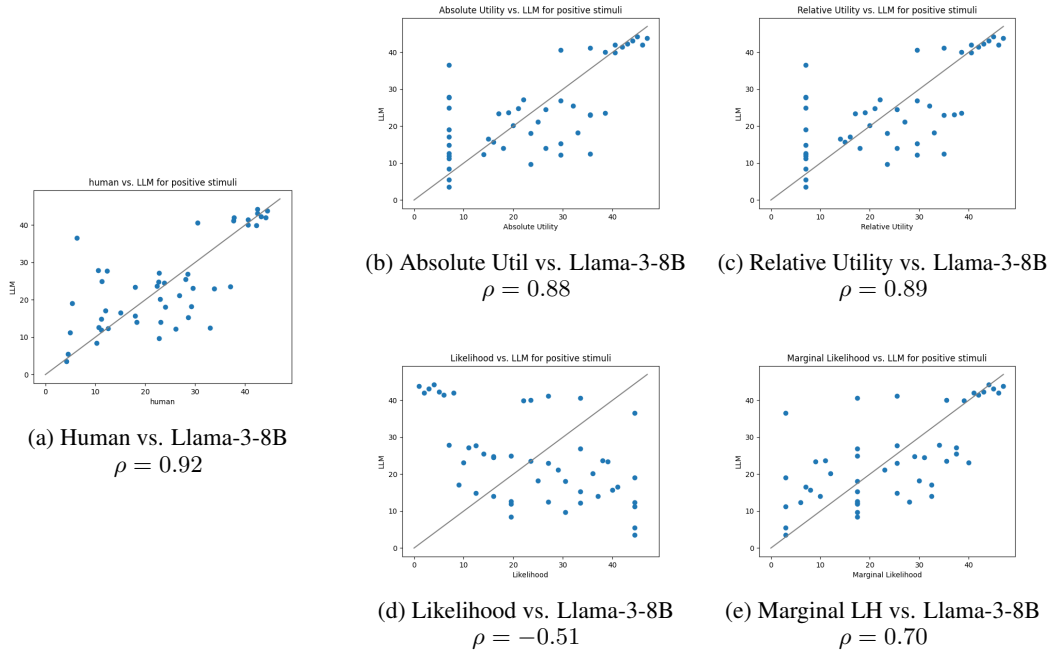


Figure 7: Comparing Llama-3-8B CoT rankings (y-coordinates) to humans and four theoretical decision-making models (x-coordinates) in negative setting.

Table 16: Example prompt for inverse modeling, zero shot.

Inverse Modeling, zero-shot:

The following are two choices that people have made between different bags of candy. Each candy is a different color.

Choice 1 was made between the following bags:

Bag 1: red, brown, yellow, blue.

Bag 2: black.

The person making the choice chose Bag 2.

Choice 2 was made between the following bags:

Bag 1: yellow, black, red, brown.

The person making the choice chose Bag 1.

People were required to choose among the bags available, and were not allowed to reject all the bags.

For example, when there is only one bag, the person has no choice but to choose it.

Which choice (1 or 2) more strongly suggests that the person making the choice likes black candies?

Please respond with either "Choice 1" or "Choice 2". Do not include anything else in your answer.